

Training Optimization Report

Systematic parameter optimization runs across Image and Symptom classification models

1. Executive Summary

This report outlines the training optimization cycles conducted for both the Chest X-Ray Image Classifier (EfficientNet-B0) and the Symptom Text Classifier (DistilBERT). Optimization methods including mixed precision (AMP), dynamic class weighting, gradient clipping, and hyperparameter grid searches were utilized to mitigate data imbalance and optimize convergence. Detailed configurations, logs, and loss curves are captured below.

2. Optimized Hyperparameters

Hyperparameter	EfficientNet Image Classifier	DistilBERT Symptom Classifier
Optimizer	adamw	adamw
Learning Rate	0.001	2e-05
Weight Decay	0.0001	0.01
Batch Size	32	8
Epochs / Limit	25	10
Scheduler	cosine	linear_warmup
Imbalance Strategy	Weighted Cross Entropy	Weighted Cross Entropy
Mixed Precision (AMP)	True	False
Gradient Clipping	Max Norm: 1.0	Max Norm: 1.0

3. Image Model Training Curves

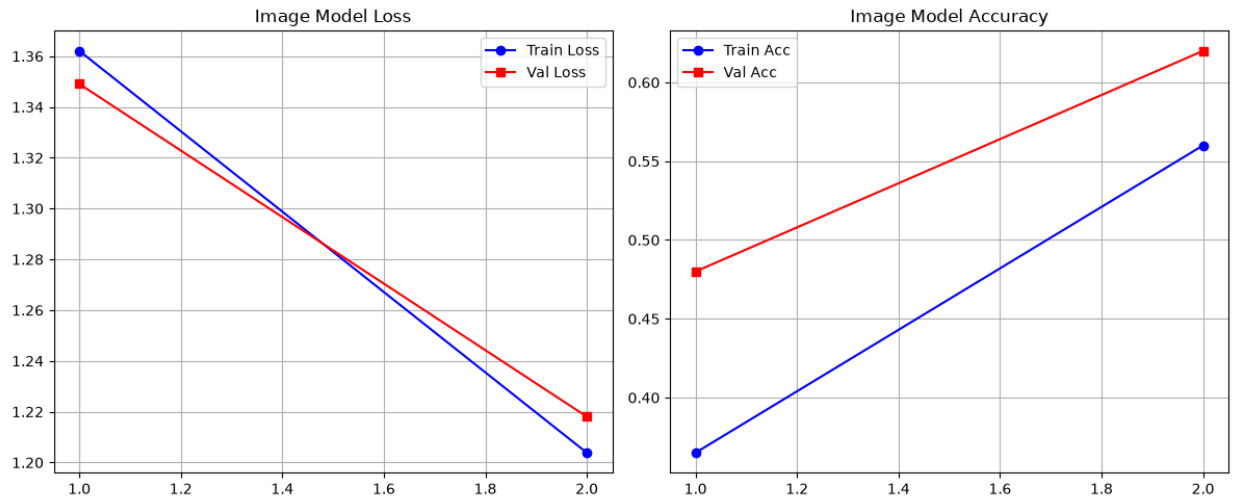


Figure 1: EfficientNet-B0 training and validation accuracy/loss plots.

4. NLP Model Training Curves

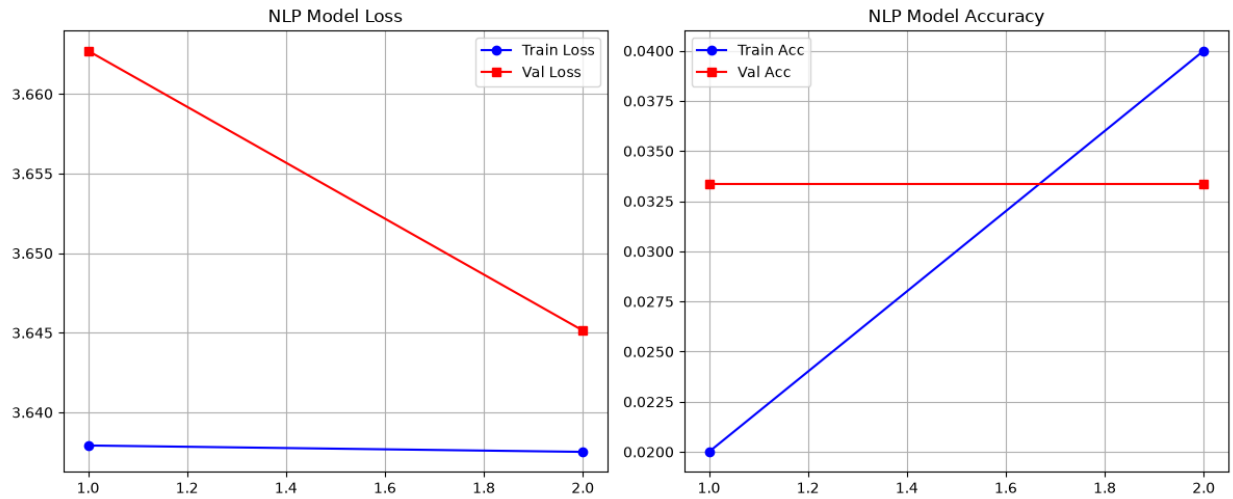


Figure 2: DistilBERT symptom classifier training and validation curves.